

Advanced Econometrics I

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Week 12 Lecture

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Extremum Estimators, M-Estimation

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Many estimators share a common structure

They are members of a broader class

This commonality can be useful for deriving consistency and asymptotic normality for a broader class of estimators

Instead of deriving asymptotic results for individual estimators, we only need to establish it once for the broader class

The class of estimators that we are looking at today are those estimators that maximize an objective function that depends on data and sample size

This class of estimators is called *extremum estimators* or *M-estimators*

Today's lecture is a summary of Newey and McFadden, *Handbook of Econometrics*, chapter 36

Wooldridge's graduate textbook also has a good discussion

The nice thing about today's lecture is that it ties together all our estimators from earlier in the semester

I will show how they can be regarded as special cases of a more general framework

We will establish consistency and asymptotic normality results for this more general framework

As a result, consistency and asymptotic normality follow for the special cases

Another nice thing of today's lecture is that it builds you a bridge to EMET8008

Thomas will teach you GMM estimation which is closely related to the framework presented here

(Unfortunately, EMET8008 won't be offered next semester)

Definition (Extremum Estimator)

Given data $W_i, i = 1, \dots, N$, an estimator $\hat{\theta}^{\text{EE}}$ is called **extremum estimator** if there is an objective function $Q_N(W_i, \theta)$ such that

$$\hat{\theta}^{\text{EE}} := \underset{\theta \in \Theta}{\operatorname{argmax}} Q_N(W_i, \theta).$$

Often the objective function is a sample average:

$$Q_N(W_i, \theta) = \frac{1}{N} \sum_{i=1}^N q(W_i, \theta)$$

Huber (1981) calls such estimators “maximum likelihood type” estimators, which justifies the next definition

Definition (M-Estimator)

$$\hat{\theta}^{\text{M}} := \underset{\theta \in \Theta}{\operatorname{argmax}} \frac{1}{N} \sum_{i=1}^N q(W_i, \theta).$$

Aside: Wooldridge defines $\hat{\theta}^{\text{M}}$ as a *minimizer* instead (irrelevant, can simply put a minus sign in front)

The class of extremum estimators contains the class of M-estimators

All our estimators from this semester satisfy this general structure

They are extremum estimators and M-estimators

$$\hat{\theta}^{EE} := \underset{\theta \in \Theta}{\operatorname{argmax}} \underbrace{\frac{1}{N} \sum_{i=1}^N q(W_i, \theta)}_{Q_N(W_i, \theta)}$$

where	OLS	$q(W_i, \theta) = -(Y_i - X_i' \theta)^2$
	IV	$q(W_i, \theta) = -(Y_i - \hat{\pi}' Z_i \theta)^2$
	ML	$q(W_i, \theta) = \ln f_Y(Y_i \theta)$

The asymptotic properties of $\hat{\theta}^{EE}$ depend on the limit behavior of $Q_N(W_i, \theta)$

Using the WLLN, let's look at the probability limit of $Q_N(W_i, \theta)$

Denote it by $Q_0(\theta)$: $Q_N(W_i, \theta) = Q_0(\theta) + o_p(1)$

In our applications:

$$\text{OLS} \quad Q_0(\theta) := -E(Y_i - X_i' \theta)^2$$

$$\text{IV} \quad Q_0(\theta) := -E(Y_i - \pi' Z_i \theta)^2$$

$$\text{ML} \quad Q_0(\theta) := E(\ln f_Y(Y_i | \theta))$$

While we have pointwise convergence at all $\theta \in \Theta$ by the WLLN, for a generic consistency proof we need something stronger...

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The class of extremum estimators and M-estimators is large

OLS, IV and ML are fairly simple cases

Here more complicated objective functions that are also covered within this more general framework:

- least absolute deviations (LAD) estimator

$$\hat{\theta}^{\text{EE}} := \operatorname{argmax}_{\theta} \left(-N^{-1} \sum |Y_i - X_i' \theta| \right)$$

- generalized method of moments (GMM) estimator

$$\hat{\theta}^{\text{EE}} := \operatorname{argmax}_{\theta} \left(- \left[N^{-1} \sum g(W_i, \theta) \right]' \hat{\Sigma} \left[N^{-1} \sum g(W_i, \theta) \right] \right),$$

where g is a *moment function* such that $E(g(W_i, \theta_0)) = 0$,
and $\hat{\Sigma}$ is a psd weight matrix that can be chosen efficiently

Aside:

letting $g(W_i, \theta) := Z_i \cdot (Y_i - X_i' \theta)$ covers the 2SLS estimator

Our goal is to obtain general conditions under which such a broad collection of estimators will be consistent

Good initial guess: convergence in probability of objective function:

$$Q_N(\theta) = Q_0(\theta) + o_p(1)$$

This guarantees that, for every θ , Q_N is close to Q_0 when N is large

The above statement is a sort of pointwise convergence result

Unfortunately, for some pathological cases, pointwise convergence isn't sufficient to establish consistency of the estimator

The next slide gives an example (feel free to ignore)

Example in which pointwise convergence is not enough

Let $Q_0(\theta) = -\theta^2$ for θ in $[0, 1]$

Clearly, $\operatorname{argmax} Q_0 = 0$

Here's a sample objective function that converges pointwise:

$$Q_N(\theta) = -\theta^2 + 2 \cdot \mathbb{1}\left(\theta = \frac{1}{2} + \frac{1}{N}\right) = Q_0(\theta) + 2 \cdot \mathbb{1}\left(\theta = \frac{1}{2} + \frac{1}{N}\right)$$

Fix any value for θ and see that the indicator term will always evaluate to zero for sufficiently large (yet finite) N

Therefore, $Q_N(\theta) \rightarrow Q_0(\theta)$ for every θ (pointwise convergence)

However, notice that $\operatorname{argmax} Q_N = \frac{1}{2} + \frac{1}{N} \neq \operatorname{argmax} Q_0 = 0$

Therefore $\hat{\theta} = \lim_{N \rightarrow \infty} \frac{1}{2} + \frac{1}{N} = \frac{1}{2} \neq 0$

A stronger convergence requirement rules out such pathological cases ...

Definition (Uniform Convergence in Probability)

A sequence $Q_N(W_i, \theta)$ is said to **converge uniformly in probability** to $Q_0(\theta)$ if

$$\sup_{\theta \in \Theta} |Q_N(W_i, \theta) - Q_0(\theta)| = o_p(1).$$

Uniform convergence implies pointwise convergence, not vice versa

Theorem (Consistency of Extremum Estimators)

If there is a function Q_0 such that

- (i) Q_0 is uniquely maximized at θ_0 ;*
- (ii) Θ is compact;*
- (iii) Q_0 is continuous;*
- (iv) $Q_N(W_i, \theta)$ converges uniformly in probability to $Q_0(\theta)$;*

then $\hat{\theta}^{EE} = \theta_0 + o_p(1)$.

Rough idea behind the previous result:

- uniform convergence, roughly, means that Q_N is very similar to Q_0 when N is large
- by definition then, $Q_N(\hat{\theta}^{EE})$ will be close to $Q_0(\theta_0)$ (because Q_0 is uniquely maximized at θ_0)
- continuity of Q_0 suggests continuity of Q_N in the limit
- This ensures that when $Q_N(\hat{\theta}^{EE})$ is close to $Q_0(\theta_0)$, then $\hat{\theta}^{EE}$ must be close to θ_0

Sketch of proof: consistency

(to get a rough idea how these four conditions are used)

- uniform convergence, roughly, means that Q_N is very similar to Q_0 when N is large
- $\hat{\theta}$ as the maximizer of Q_N is then also a maximizer of Q_0
- but then, $Q_0(\hat{\theta}) > Q_0(\theta_0) - \varepsilon$, for every positive ε
- this sort of shows that $\hat{\theta}$ is close to θ_0
(more details on next slide, if you're interested)

- pick any closed subset of Θ not containing θ_0
(remove an open set containing θ_0 from Θ , the set that remains is compact because Θ is compact)
- by (iii) we can use Weierstrass' theorem:
a continuous function over a compact set attains a maximum
- denote this maximum (over the compact subset) by $Q_0(\theta^*)$
- by (i): $Q_0(\theta_0) - Q_0(\theta^*) > 0$
- from previous slide: $Q_0(\hat{\theta}) > Q_0(\theta_0) - \varepsilon$, for every positive ε
- set $\varepsilon = Q_0(\theta_0) - Q_0(\theta^*)$, therefore $Q_0(\hat{\theta}) > Q_0(\theta^*)$
- then $\hat{\theta}$ must be from the open set containing θ_0
- it follows that $\hat{\theta} \xrightarrow{p} \theta_0$

(for details consult proof of Theorem 2.1 in Newey and McFadden)

The theorem gives four conditions

How should we interpret them?

Distinguish *substantive* conditions from *regularity* conditions

Conditions (i) and (ii) are substantive, meaning they could be restrictive in some applications

Conditions (iii) and (iv) are regularity conditions, meaning they are satisfied in many applications

In addition, we can think of conditions to be *primitive* or *non-primitive*

A condition is called primitive, if it is easy to interpret (akin to high-level programming languages)

Only one of the four conditions is primitive: the compactness condition

In practice, condition (ii) is rarely checked

If you are estimating a probability, then compactness is easily verified

But if you run a regression, then you are implicitly thinking that your coefficients come from a closed and bounded set

Condition (i) is substantive and non-primitive

It has to be checked on a case by case basis

For the maximum likelihood model, we actually have checked condition (i) already

Recall from the week 10 lecture, that for $\theta \neq \theta_0$,

$$E(\ln f(Y|\theta_0)) > E(\ln f(Y|\theta))$$

This means that the expected value of the log likelihood is maximized at the *true* value of the parameter

In other applications, condition (i) may not hold

What about conditions (iii) and (iv)?

They are non-primitive regularity conditions

While they are not easy to interpret, we believe that they may be easily satisfied nevertheless

The following Lemma helps when Q_N is based on a sample average, that is, when we consider M-estimators

Lemma (Uniform Law of Large Numbers for M-Estimators)

Let $Q_N(W_i, \theta) = \frac{1}{N} \sum_{i=1}^N q(W_i, \theta)$ and $Q_0(\theta) = E(q(W_i, \theta))$. If

- (i) the sample data W_i are iid;
- (ii) Θ is compact;
- (iii) $q(W_i, \cdot)$ is continuous at each $\theta \in \Theta$ with probability one;
- (iv) there exists a dominating function $d(W_i)$ with $E(d(W_i)) < \infty$, such that $|q(W_i, \theta)| \leq d(W_i)$ for all $\theta \in \Theta$;

then

- $Q_N(W_i, \theta)$ converges uniformly in probability to $Q_0(\theta)$, and
- $Q_0(\theta)$ is continuous.

The conditions in the Lemma are all primitive and also quite weak

Notably, $q(W_i, \cdot)$ can be discontinuous on a set of measure zero

Conditions (iii) and (iv), both on q , are reasonably easy to check

Condition (iii) can be checked by inspection and condition (iv) boils down to existence of moments

OLS example (scalar X_i)

- let β be from the compact set $[\underline{b}, \bar{b}]$
- $q(W_i, \theta) = -(Y_i - X_i\theta)^2$, clearly continuous
- dominating function

$$\begin{aligned}|q(x, y, b)| &:= (y - xb)^2 = 2y^2 + 2b^2x^2 - (y + xb)^2 \\ &\leq 2y^2 + 2b^2x^2 \\ &\leq 2y^2 + 2\bar{b}^2x^2 =: d(x, y)\end{aligned}$$

- notice that $E(d(X_i, Y_i)) = 2E(Y_i^2) + 2\bar{b}^2 E(X_i^2) < \infty$
if second moments of X_i and Y_i are finite

Connecting the dots:

To establish consistency of an M-estimator, we need to check the four conditions of the Lemma plus the identification condition (condition (i)) of the theorem

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For the class of extremum estimators, there also exists a generic result on the asymptotic distribution

Conditions to establish asymptotic normality are a bit involved

However, if we restrict attention to extremum estimators with objective functions that are sufficiently smooth, then we can use an intuitive approximation via the mean value theorem

All we need is for Q_N to be twice continuously differentiable

Sketch of proof: asymptotic normality

(this is similar to the asymptotic distribution of MLE in week 10)

Notice, by definition, $\nabla_{\theta} Q_N(W_i, \hat{\theta}^{EE}) = 0$

Apply the mean value theorem around θ_0

$$0 = \nabla_{\theta} Q_N(W_i, \hat{\theta}^{EE}) = \nabla_{\theta} Q_N(W_i, \theta_0) + (\nabla_{\theta\theta} Q_N(W_i, \tilde{\theta})) (\hat{\theta}^{EE} - \theta_0),$$

where $\tilde{\theta}$ is between θ_0 and $\hat{\theta}^{EE}$

Multiplying through by $\sqrt{N}$ and solving for $\sqrt{N}(\hat{\theta}^{EE} - \theta_0)$,

$$\begin{aligned} \sqrt{N}(\hat{\theta}^{EE} - \theta_0) &= -(\nabla_{\theta\theta} Q_N(W_i, \tilde{\theta}))^{-1} \cdot \sqrt{N} \nabla_{\theta} Q_N(W_i, \theta_0) \\ &\quad \downarrow_p \text{ (WLLN)} \qquad \qquad \qquad \downarrow_d \text{ (CLT)} \\ &= H^{-1} \cdot \mathcal{N}(0, \Sigma) \\ &\stackrel{d}{\rightarrow} \mathcal{N}(0, H^{-1} \Sigma H^{-1}) \end{aligned}$$

(also using Slutsky's theorem)

The formal result is

Theorem (Asymptotic Normality of Extremum Estimators)

Let $\hat{\theta}^{EE}$ be an extremum estimator such that $\hat{\theta}^{EE} = \theta + o_p(1)$. If

- (i) $\theta_0 \in \text{interior}(\Theta)$;
- (ii) $Q_N(W_i, \theta)$ is twice continuously differentiable in a neighborhood T of θ_0 ;
- (iii) $\sqrt{N} \nabla_{\theta} Q_N(\theta_0) \xrightarrow{d} \mathcal{N}(0, \Sigma)$;
- (iv) there exists $H(\theta)$ that is continuous at θ_0 and
$$\sup_{\theta \in T} |\nabla_{\theta\theta} Q_N(W_i, \theta) - H(\theta)| = o_p(1)$$
- (v) $H := H(\theta_0)$ is nonsingular;

then $\sqrt{N}(\hat{\theta}^{EE} - \theta_0) \xrightarrow{d} \mathcal{N}(0, H^{-1}\Sigma H^{-1})$.

Condition (i) is necessary because estimators may not be asymptotically normal when θ_0 sits at the boundary of Θ (this is not easy to show)

Condition (ii) is stronger than necessary, but convenient to obtain the Hessian matrix as the probability limit

Condition (iii) is easy to establish for M-estimators with differentiable q (such as OLS, IV, and MLE)

Condition (iv) is a U-WLLN for the Hessian

Condition (v) makes sure that the Hessian is invertible

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For M-estimators in which q is differentiable, it is easy to estimate the asymptotic variance $H^{-1}\Sigma H^{-1}$

You can simply use plug-in estimators for H and Σ

Consistent estimators are

$$\hat{H} = \nabla_{\theta\theta} Q_N(W_i, \hat{\theta}^{EE}) = (1/N) \sum_{i=1}^N \nabla_{\theta\theta} q(W_i, \hat{\theta}^{EE})$$

$$\hat{\Sigma} = (1/N) \sum_{i=1}^N \nabla_{\theta} q(W_i, \hat{\theta}^{EE}) \nabla_{\theta} q(W_i, \hat{\theta}^{EE})'$$

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Final Exam

Three questions, each counting 20 raw marks (will be deflated by $5/6$ when calculating your overall course mark)

- Question 1: analytical question taken straight from problem sets 5, 6, 7, 8, 9, or 10
- Question 2: four true/false questions (each counting 5 marks)
- Question 3: new analytical question, part of which can be answered by directly applying what you've already learned; the other part asks you to adapt that knowledge to an unfamiliar or modified context

To make your lives a little easier during the next few weeks, here are two topic/areas that will NOT be asked in the final exam:

- weak instruments
- extremum estimators, M-estimators (as in week 12)

Additional Consultations

To answer any remaining questions that you may have and to support your learning towards the final exam, we are going to offer the following additional consultation sessions:

- Tuesday June 2 from 10-11am @ Arndt 1022
- Tuesday June 9 from 10-11am @ Arndt 1022
- Monday June 15 from 10-11am @ Arndt 1022

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Last Slide (yey!)

I hope you ...

- enjoyed it a bit
(didn't hate it too much!?)
- learned something useful

Feel free to come by my office for a chat anytime!

Good luck with all your exams!